the CPU could suffer from surges or spikes which could either burn the processor or shut it down temporarily. The simple rule is that the more phases there are the better regulated your power supply is although anything above 12 phases is overkill for the majority of computer users. Overclocking is one feature that heavily depends on a stable power supply. Power phase design follows a simple notation where it is written as x + y +z or x + y. Example, 8+2 power phase denotes that 8 MOSFETs regulate the power for your CPU and the remaining two is divided over your RAM and memory controller / Integrated graphics.

The reason for their existence is that CPUs require a lot of power and having a single source of power results in a lot of heat being dissipated by that one component, heat is bad for anything that has to do with electricity. So in order to keep temperatures lower you have multiple sources of power, this way each phase remains on for a short duration / duty cycle and then the next one turns on while the previous switches off. This way you are ensured of a steady power supply that does not vary much. The more phases there are the lesser amount of heat that each VRM component has to dissipate. Naturally, high load operations like overclocking benefit by a huge margin with a good power phase design. While the notation used is not the absolute standard, the motherboard packaging should mention the power phase design it has. Even graphics cards have a VRM circuitry.

5. BIOS

The first thing that comes to life when you switch on your computer is the BIOS (Basic Input/Output System). The primary objective is to get the computer's operating system started but before that happens the BIOS does quite a lot of other things too, they are listed as follows -

- A POST (Power-On Self Test) where each essential component is tested once to check if there are any faults. If any are found then a sequence of beeps are issued to let you know about it. Newer motherboards have an LED display which displays error codes. You can refer to your motherboard manual to figure out which code means what.
- Switching on the BIOS on other peripherals. Every add-on card has it's own BIOS, GPUs, Ethernet Cards etc have their own BIOS which is switched on by the main BIOS.
- The initial management of input devices like keyboard and mice is done by the BIOS prior to the OS booting up.